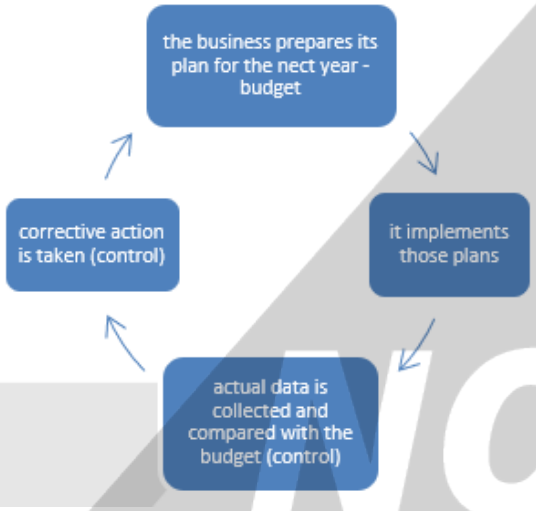
1. Elements of costing and management accounting

1.1. Costing for materials, labour and overheads

- Cost accounting is a method of accounting where all costs relating to a certain activity are grouped together, classified and recorded
- Management accounting is the process of preparing reports and accounts to be used by internal parties such as managers for decision making on future performance of the business
- Direct costs are costs incurred with directly relate to the production of the good or service example: raw materials, carriage inwards, royalties.
- Indirect costs are costs that aren't directly linked with the production of goods and services example: cleaning materials, supervisor salary, factory managers.
- Fixed costs are costs that remain unchanged while the level of output or activity increases or decreases, example: rent
- Stepped fixed cost is cost that does not change within certain high or low threshold of the activity but which will change once these thresholds are breached



- Variable costs are costs that varies on the level of output of activity, example: direct costs that vary in proportion to the number of units produced
- Cost unit is the unit of output of a business to which costs can be charged
- IAS 2 allows two different methods for valuing direct material inventory.

1.2. Cost accounting

- Enables managers to manage their business effectively and efficiently.
- Used by all types of businesses as all types of business organisations need to know the total costs involved in running the business.

Cost Classification

- Costs may be classified in one of three ways:
  1. By element
  2. By function
  3. By nature

Classification by element

- In a manufacturing organisation, there are three elements to cost.
  1. Material – cost of goods used
  2. Labour – the wages and salaries cost
  3. Expenses – all other expenses, including overheads.
- Each of these elements may be split into :
  - Direct costs – those costs that can be specifically associated with the manufacture of one unit of production (e.g., raw material, wages paid.)
  - Direct costs are the prime cost of manufacture.
  - Indirect costs are those costs that cannot be specifically associated with the manufacture of one unit of production. (e.g., Salaries, rent, light, and heat)
  - The indirect costs of a business are called overheads.

Classification by function

- Production
  - The cost of production is calculated as: Opening inventory + Net purchases + Carriage inwards – Drawings of inventory – Closing inventory.
- Sales and distribution: Expenses related to the transportation/distribution and the sales made by the business.
- Administration: Expenses related to the administration of the business
- Finance: interest, taxes, etc. Those that are charged to business finances
- NOTE: Charges, in this case depreciation, can be divided among two of the classifications, these divisions will be clearly stated in the paper. At Least one classification will be addressed in the question to charge depreciation to.

Classification by nature

- Fixed costs: Those costs that do not vary with the levels

of business activity. Eg- rent, salaries, insurance.

- Variable cost: those costs that change in direct

proportion to changes in the levels of activity. Eg- direct

material, piece rate labor cost.

- Semi variable cost: It has characteristics of both fixed

cost and variable cost. Eg – telephone bill.

- Stepped cost: remain fixed until a certain level of activity

is reached, at which they rise to a higher fixed level. Eg –

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supervisor salaries – 20 employees, from 21st employee

further supervisor required.

The valuation of inventory may be difficult because goods are purchased over different time periods and different prices will be paid for the same items. There a three methods of valuing inventory:

- FIFO – First in first out
- AVCO – average cost
- LIFO – Last in first out.

1.3. FIFO

This method assumes that the items purchased first were also sold first.

Any unsold items are considered to be those purchased most recently.

Advantages of FIFO

- Closing inventory is based on the most recent prices paid.
- Inventory valuations are based on actual prices paid for items.
- It is logical because it assumes that sales are made from the oldest items of inventory first.
- Using FIFO March: 1000 units of inventory purchased at $4

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July: 500 units of inventory purchased at $8

September: 1100 units of inventory sold
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DATE	RECIEVED		ISSUED		INVENTORY	VALUATION		
	QTY	PRICE	VALUE	QTY	PRICE	VALUE	QTY	PRICE VALUE TOTAL
	$	$	$	$	$	$	$	
March	1000	4	4000		1000	4	4000	4000
July	500	8	4000		1000	4	4000	
				500	8	4000	8000	
September		1000	4	4000	400	8	3200	3200
		100	8	800				

AVCO

- The average cost method of inventory valuation involves recalculating the average cost of inventory after each new purchase.
- Formula: $\frac{\text{Current inventory value} + \text{Value of inventory purchased}}{\text{Number of units in inventory}}$

Advantage of AVCO

- Identical items in the inventory are given identical cost.
- Using the same numerical example for AVCO would look like this

Date	RECIEVED	ISSUED	Avg. Cost	~~INVENTORY~~	~~VALUATION~~	
	QTY	PRICE	VALUE	QTY	QTY	VALUE
	$	$		$		
March	1000	4	4000	4	1000	4000
July	500	8		5.33	1500	8000
September		1100	5.33	400	2132	

- *Note*: Avg. cost of inventory only changes when goods are purchased not sold. The goods are sold at the average price. When new inventory is purchased it changes the average cost of the held inventory, therefore causing the change.

LIFO

- The last in first out method assumes that the items purchased most recently were sold first.

The result of this is that any unsold items are considered to be those purchased earliest.

The effect of inventory valuation on the financial statements:

- The cost of sales when using FIFO method will be lower than when using AVCO.
- The profits using FIFO will be higher than when using AVCO
- In long term, profits using both methods will be exactly the same, as eventually all inventory will be sold, but in the short term, FIFO produce higher profits than AVCO.
- FIFO also produces higher current assets and net assets on the statement of financial position than AVCO.

1.4. Absorption Costing

Apportionment of overheads to cost centres.

- Cost centres are any location in a department to which costs may be attributed.
- Production cost centres are locations which are directly involved in producing goods
- Service cost centre are locations which are not involved in the production of goods but provide services for the production cost centres eg, stores.
- Allocation of costs is charging costs directly to the cost centres which can be directly identified with them, for instance wood, metal, plastic, components and chemicals are allocated to the production department. Packing materials are allocated to the packing department.
- Apportionment of cost is the process of charging costs to cost centres using a suitable basis, these costs cannot be directly identified with a single cost centre.

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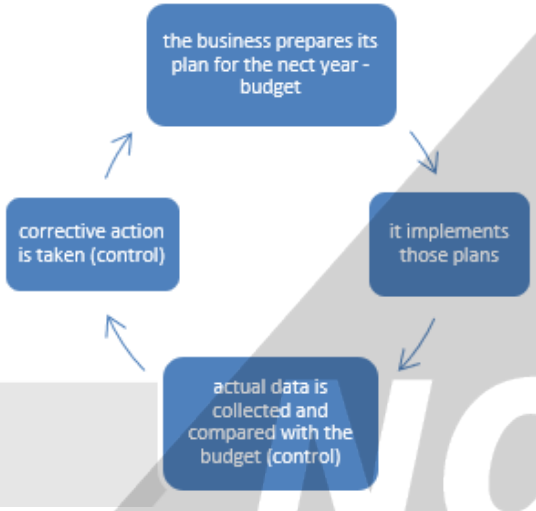
Overhead	Basis of apportionment
Rent and rates, heating and lighting, building depreciation, insurance, building expenses, rent	Floor area
Actual consumption	Power
Depreciation, insurance of plant and other machinery	Cost or replacement value of asset
Storekeeping costs, material handling costs	Store requisition
Welfare costs, canteen costs, administration costs, supervision	No. of employees
Machines repairs	Machine hours

- Steps to absorption costing
  - if a manufacturer has four production cost centres; moulding, machining, painting and packaging, Given basis for each department for rent is floor area of $1000m^2, 700m^2, 500m^2 and 300m^2$ respectively. If rent was 45000 dollars, the cost will be apportioned as below

Expense	Moulding	machining	Painting	Packaging	
	Rent	18000	12600	9000	5400

- This is known as primary apportionment Overhead analysis sheet



- After apportionment, If the service cost centres provided services to the production cost centres they will need to be reapportioned to the production centres on suitable bases, the simplest and quickest method is the elimination method (this is where the service costs are apportioned using bases which don't involve the service sot centre in question).
- Another method is the reciprocal method as shown below where the cost centres are apportioned until they reach immaterial values

	Assembly $	*Finishing* $	*Maintenance* $	*Canteen* $	*Total* $
Total from above	17,350	23,970	18,600	6,600	66,520
Reapportion canteen	2,200	3,300	1,100	(6,600)	–
Reapportion maintenance	9,850	7,880	(19,700)	1,970	–
Reapportion canteen	657	985	328	(1,970)	–
Reapportion maintenance	164	131	(328)	33	–
Reapportion canteen	11	17	5	(33)	–
Reapportion maintenance	3	2	(5)	0	–
Total	30,235	36,285	0	0	66,520

- Calculating the Overhead Absorption Rate [OAR] : rate at which costs are apportioned to a cost centre are charged to the cost unit passing through it, OARs are used to calculate overhead attributed per unit, it is calculated by:
  - budgeted production overheads/budgeted activity level (labour hours or machine hours)
- Usually both direct labour and machine hours are given, always choose the greatest and **never** add the two sets together
- Once the OAR rate is found this will be multiplied by the number of hours used to make the product to obtain the amount of overhead absorbed.
- Under/over absorption is calculated by:

Actual overheads	XXX
(less) absorbed overhead (OAR * actual hours)	(xxx)
Under absorbed	xxx

- If absorbed overheads was however greater than actual overheads then this will result in over absorbed
- Reasons for under absorption
  - Actual expenditure is greater than budgeted
  - Production is less than the planned level, not enough overheads has been charged to production.
- Reasons for over absorption
  - Actual expenditure is less than budgeted expenditure
  - Actual production is more than the planned level, too much overhead was charged to production.
- Benefits of absorption costing
  - Acceptable under IAS as well as GAAP
  - Accounting for all production cost not just direct cost hence giving a complete picture of the cost per unit, more realistic selling price as a result
  - Gives a better view on profitability then variable cost especially if not all products are used during the same period.
  - Better suited for companies which have constant demand for products

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- Drawbacks of absorption costing
  - As this is based on both fixed and variable costs, this method cannot be used for management decision making or planning further actions like budgeting
  - Absorption makes It difficult to do cost volume profit analysis, as with additional fixed cost, the variations in the variable cost becomes difficult to determine. Making it difficult to assess efficiency and effectiveness.
  - Profits are needed to be adjusted for over and under absorption of fixed costs, this can make the calculation of profits difficult and complicated especially when the company has too many products.

1.5. Unit, Job and Batch Costing

- Continuous operations are typically those in which a single type of good is produced and the cost units are identical. Production may involve a sequence of continuous or repetitive operations such as oil refining. Continuous costing is used to find the cost of a single unit of production[unit costing=total cost/output]
- Specific order operations are those which are performed in response to special orders received from customers, this may be job or batches
- Job costing is a costing method that calculates the cost of meeting a specific customer order or job such as re painting a house

A job quotation may look like

Job no.282

Installation of central heating at xxx

	Estimated costs $	Actual cost $
Materials	2800	2670
Labour	1800	2150
Overhead	3500	3380
Total cost	8100	8200
Add:		
profit 25% of cost	2025	1925
Amount of quotation	10125	10125

- Once the quotation is accepted the costs will be recorded on the job card as incurred, the actual costs may however differ from the quoted amount, in that case the business is only charging the quoted selling price hence causing profit to either increase or decrease
- Job cards would provide useful information the next time the business receives a request for a similar job, directors can identify areas of overspend and take corrective action for future jobs since they are focused on a specific area.
- Batch costing is where the costing procedure to find the cost of a number of identical items produced
- Similar to job costing the selling price on the quotation will be agreed and fixed, if everything goes as per plan the same profit may be made however if costs prices or labour hours deviate this might cause profits to fall
- This allows the business to plan for its future.

Costs for a batch of 500 chairs

	$	$
Direct materials		X
Direct labour : Machining	x	
Finishing	x	
Assembly	x	
Prime cost		x
Production overhead recovered (no of hours * OAR)		x
Machining		x
Finishing		x
Assembly		x
Cost of production		x
Add: administration costs		x
Total cost of batch of 500 chairs		
Profit		x
Selling price		xx

1.6. Marginal costing

- Marginal costing is the cost of making an extra unit of output.
- Contribution is the difference between the selling price and the variable cost of a unit of output.
  - selling price - variable cost per unit
- Contribution to sales ratio (C/S ratio) is the ratio of contribution to the selling price, calculated by;
  - $\frac{contribution}{sales} \times 100$
- Break even point is the level of output in which the business makes neither a profit nor a loss, total contribution is exactly equal to total fixed costs.
  - $\frac{total\ fixed\ costs}{contribution\ per\ unit}$ to find the breakeven sales this is multiplied by the selling price or the following equation can also be used; $\frac{total\ fixed\ cost}{\frac{c}{s}\ ratio}$
- If there the business has a target profit set the breakeven level of output will be calculated by:
  - $\frac{fixed\ cost + target\ profit}{contribution\ per\ unit}$

Breakeven charts

- Are diagrammatic representation of profit or loss to be expected from the sale of a product at various levels of activity.

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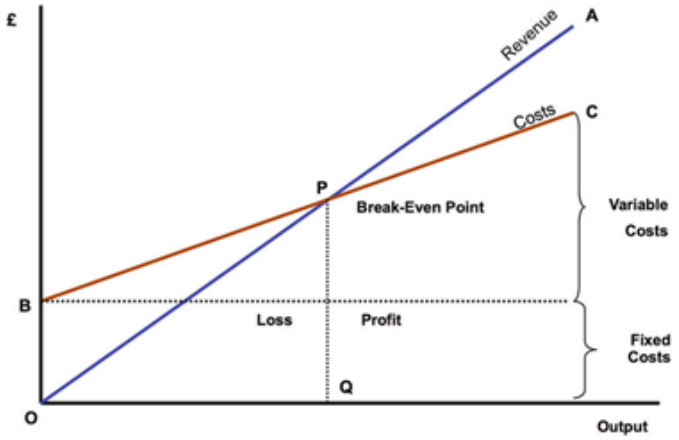
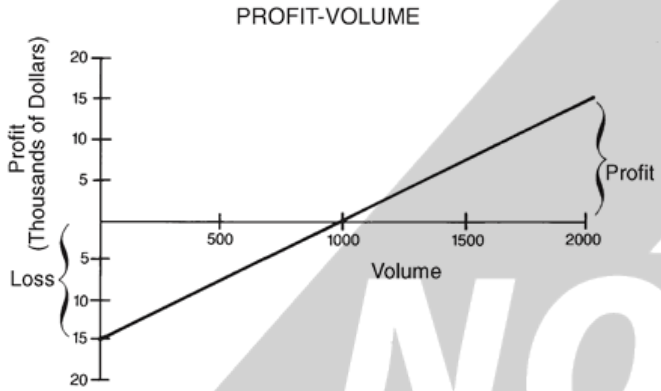


- Cost volume profit analysis is a way of finding out how changes in variable and fixed cost affect a firms profit. These are used by companies to see how many units they need to sell to breakeven or reach a certain minimum profit margin.
- Margin of safety is the difference between the breakeven point and the level of sales , this is found by total units sold - breakeven level of output or the following equation is used to calculate margin of safety rate $\frac{profit}{ratio}$
- Profit volume chart is a type of breakeven chart which only shows the profit of loss at each level of output.
- At 1000 units of output the business incurs neither a profit nor a loss, the units exceeding the BEO is the margin of safety which is 1000 units.



Limitations of breakeven charts

- it assumes there are no changes in the level of inventory, everything produced is assumed to be sold during that period which is unrealistic.
- It doesn't allow for a mix of products and is usually calculated for a single product which.
- Cost behaviour is assumed to be either fixed or variable, so semi variable costs are not considered. Similarly stepped costs are also not included
- Variable costs are assumed to be perfectly linear with the level of production so changes in costs such as bulk purchase discounts or overtime rates aren't considered.
- The selling price is assumed to remain fixed throughout the year so seasonal sales or discounts aren't considered, this is misleading.

Advantages of breakeven analysis

- Charts are easy to construct and understand
- Provides useful guideline for managers like breakeven points, safety margins when taking decisions such as on ordering raw materials and budgeting.
- Comparison between different options and draw a new chart to show changing situations such as a price increase.

Acceptance of orders below normal selling price

- When the order results in a contribution to cover fixed costs, the selling price must exceed marginal cost of production
- To maintain production and avoid laying off skilled workforce during off season
- To promote a new product.
- To dispose of slow moving or redundant inventory
- The business should consider if they have the available spare capacity.

Make or buy decisions

- if the marginal cost of production is below the price quoted by the supplier then the offer should be rejected.
- The business should consider
  - Will the price offered by the suppliers change?
  - The quality of the supplies offered by the external supplier.
  - Will there be delays in delivery?
  - Is there an alternative use for the resources?
  - Will the business lose the services of skilled workers which may be difficult to replace at a later date?

Dropping a product

- if a business is dealing with a range of products and one of which is deemed to be unprofitable, it may consider dropping the item from its range.
- When the contribution is negative, but if the product has a positive contribution and a negative profit, it is favourable to continue the production as it would still be covering some of the fixed cost.

Choice of product where a limiting factor exist

- a limiting factor is anything that limits the quantity of goods that a business may produce. This may include the shortage of materials, labour and demand for a particular product.

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- When making the best use of limited resources the product must be ranked according to the amount of contribution they make from each unit of the scarce resource.
  - Step 1: calculate the contribution per unit
  - Step2: identify the limiting factor and limiting factor per unit
  - Step 3: find $\frac{contribution}{limiting\ factor\ per\ unit}$
  - Step 4: based on the answer, state the rank in descending order
  - Step 5: calculate the optimum production capacity, produce a revised production plan considering the shortage

Marginal costing VS absorption costing

- Marginal costing: using this method, fixed costs are treated as period costs and written off each period. Finished goods and work in progress are valued at marginal cost only i.e. variable costs.
- In inventory valuation, when using marginal costing the closing inventory will only include the variable cost. While, in absorption the variable as well as some of the fixed overhead will be included hence giving a greater gross profit for absorption as closing inventory is valued more than marginal. This difference arises purely due to the way the fixed overheads are treated.

Format of an operating statement (marginal costing)

	$	$
Sales	x	
(-) variable cost:		
Materials	x	
Labour	x	
Overheads	x	
(+) opening inventory	x	
(-) closing inventory \n $\frac{total\ variable\ production\ cost}{total\ units\ produced} \times closing\ inventory$	(x)	(x)
	x	
(-) selling variable expenses	(x)	
Contribution	x	
(-) fixed costs		
Admin costs	x	
Production costs	x	(x)
Profit for the year	X	

- Absorption costing uses all costs which are absorbed into production and thus operation statements do not distinguish between fixed and variable cost. Valuation of both finished and work in progress inventory contain both fixed and variable costs.

Format of an operating statement (absorption)

	$	$
Sales	x	
(-) production costs		
materials	x	
labour	x	
fixed overheads	x	
(+) opening inventory	x	
(-) closing inventory $\frac{total\ variable\ production\ cost}{total\ units\ produced} \times closing\ stock\ units$	(x)	x
gross profit	x	
(-) fixed and variable operating expenses		
admin expenses	(x)	
selling and distribution expenses	(x)	
profit for the year	x	

Uses of marginal costing

- Clear distinction between variable and fixed costs which is helpful in short term decision making.
- Sees costs allocated to a time period so that it may be argued that profits for the time period is more accurate.
- It is prudent to write off costs in the time period incurred and therefore marginal costing methods include accurate inventory valuation of and as well as profits.
- Easy to operate but relied upon cost being split into fixed and variable. No arbitrary apportionment of fixed costs.
- Shows clearly the impact on profit of fluctuations in the volume of sales.
- Under and over absorption is not a problem hence no need to calculate and OAR
- Can also be used with standard costing.
- Shows relative contributions to profit that are made by each product and shows where the sales effort should be contracted.

Disadvantages of marginal costing

- Not all costs can be split into variable and fixed
- Fixed costs don't in reality remain constant and variable costs do not vary according to levels of output, in the long run all costs are variable
- Assumes production method remains constant which isn't the case in reality.
- Contribution may vary if new techniques are followed in the production process.
- It isn't useful for business with a range of products as allocating fixed costs to each product will be difficult with any degree of accuracy, making difficult to calculate the breakeven point.

Uses of absorption costing

- Includes all costs to product there it could be useful for management when fixing prices or reviewing of a product has been profitable.

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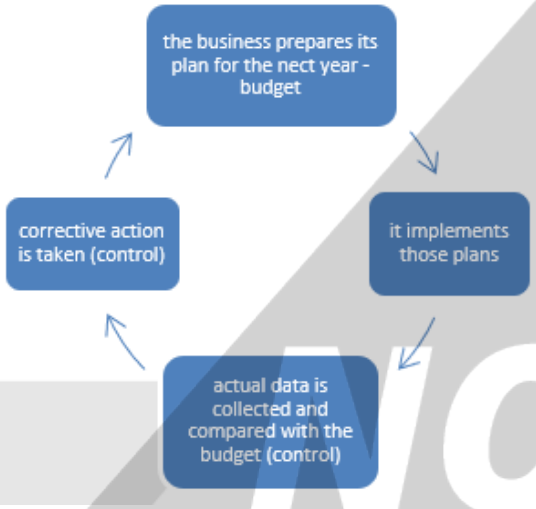
- It could be said that profit gives a true and fair view according to absorption costing
- Helps set prices.
- Used in the long run rather than short run.
- More acceptable and realistic for financial statements.

High low method

- This technique uses the units and total cost at one level of activity compared to the units and total costs incurred at another level to calculate the variable cost per unit.
- This is used to split and find the fixed element and the variable element of a semi variable costs.
- Once variable cost is found, fixed costs can be found by deducting the variable cost from the total cost.
  - $\frac{highest\ cost - lowest\ cost}{highest\ activity - lowest\ activity} = variable\ cost\ per\ unit$

1.7. The application of accounting to business planning

- A budget is a short term financial plan of standard cost and revenue prepared for the future to control resources so that the business objectives can be achieved.
- Budgeting is the preparation of the budget for each budget centre.
- The planning process includes



- Planning – managers are responsible for planning and controlling a business and budgets are an essential tool for planning and controlling
- Rolling budgets is where as each year passes it is deleted from the budget and a budget for another year is added, making one year detailed and the following years less precise. Top down budgets are prepared by top management and handed down to departmental managers. Bottom up budget is prepared by departmental managers and although they tend to reflect local conditions better, they may be set easily achievable and may not fit with overall plan.
- Managers should concentrate their attention on items with adverse variance (where the actual performance deviates negatively from the budgeted) management accounts may report only the items with adverse variance this is known as management by exception or exception reporting.

The need for budgeting

- A forecast of income and expenditure (thereby profitability) helps figure out long term goals and works towards them.
- A tool for decision making
- A means to monitor business performance
- Ensure the business doesn't spend money they don't have.
- Helps build investment contribution into your budget
- Helps the business be prepared in unforeseen instances
- Helps identify unnecessary expenditure and cut down on these
- Ensure the business has money for future projects

Budgetary control

- Refers to the responsibility that each budget centre has to justify the use of resources, control costs, achieve activities set by the budget in accordance with the business objectives.
- delegates financial planning to manage it evaluates managers performance by comparing actual results with those set in the budget.

Advantages of budgeting and budgetary control

- Facilitates long term planning, leads managers to think about the future and the direction in which the business is heading.
- Promotes coordination between department.
- Enables monitoring and control.
- Acts as a motivator for employees.
- Helps allocation and use of resources.
- May provide a framework for delegation.
- Helps in decision making.
- Helps identify areas where improvement is required before start of a financial period.
- May help in securing finance as banks and lenders requires budget information.

Disadvantages of budgeting and budgetary control.

- May demotivate staff if targets are set too challenging.
- Time consuming and costly, requires specialist knowledge.
- Can cause conflicts between department about resource allocation.
- Can discourage innovation – budgets restrict activity managers fail to take advantages of unexpected opportunities as actions are restricted.
- If managers are paid on their performance against the budget, they may build extra costs and reduce the revenue from the expected target to ensure they always beat it, this is referred to as budgetary slack.

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